



SOICT

CAPSTONE PROJECT

COMPUTER VISION – IT4343E

Fundamental Concepts and Methods in Biomedical Image Registration

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Abstract

Biomedical image registration aligns two medical images so that corresponding anatomical structures occupy the same spatial location. It is an important step in longitudinal analysis, atlas construction, label transfer, multimodal fusion, and image-guided treatment. This project studies the basic registration problem and compares four representative solution families in one runnable framework: classical optimization, metaheuristic search, convolutional neural networks, and Transformers.

The implemented methods are SimpleITK Diffeomorphic Demons, a Particle Swarm Optimization (PSO) method for three-dimensional affine registration, a compact VoxelMorph-style convolutional network, and a compact TransMorph-style Transformer network. The two learned methods predict dense displacement fields and use the same differentiable spatial transformer and unsupervised loss. The methods are evaluated on FreeSurfer-derived OASIS-1 T1-weighted brain MRI data. The experiment contains 425 subjects and 424 adjacent image pairs resized to $64 \times 64 \times 64$. A fixed 70/10/20 pair-level split gives 297 training, 42 validation, and 85 test pairs. Final metrics are reported only on the test set.

The results show that no single method is best for every objective. VoxelMorph obtains the lowest mean squared error (0.008916), the highest normalized cross-correlation (0.827261), and the fastest mean inference time among the four methods (0.3631 seconds per pair). PSO obtains the highest anatomical Dice score (0.330794), but it is much slower (9.3619 seconds per pair). TransMorph provides a better balance between Dice and deformation stability than VoxelMorph, while Classical Diffeomorphic Demons produces no detected Jacobian folding in this benchmark. These findings illustrate an important practical point: intensity agreement, anatomical overlap, runtime, and deformation regularity measure different aspects of registration quality and should be considered together.

Keywords: biomedical image registration, OASIS-1, Diffeomorphic Demons, Particle Swarm Optimization, VoxelMorph, TransMorph.

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List of Abbreviations

CNN	Convolutional Neural Network
CT	Computed Tomography
DDF	Dense Displacement Field
DIR	Deformable Image Registration
GPU	Graphics Processing Unit
MSE	Mean Squared Error
MRI	Magnetic Resonance Imaging
NCC	Normalized Cross-Correlation
OASIS	Open Access Series of Imaging Studies
PSO	Particle Swarm Optimization
STN	Spatial Transformer Network
ViT	Vision Transformer

Chapter 1

Introduction

1.1 Background and Motivation

Biomedical image registration is the process of aligning two or more medical images in a shared coordinate system. One image is called the *fixed image* and defines the target coordinate system. The other image is called the *moving image*. Registration estimates a transformation that moves or deforms the moving image so that corresponding anatomical structures match the fixed image [1, 2].

Medical images are rarely aligned automatically. The same patient may be placed in a different position during two scanning sessions. Soft tissue can deform, the head may rotate, breathing may move organs, and scanner settings may change. For images from different people, normal anatomical variation creates an even larger difference. Registration provides a systematic way to reduce these spatial differences before further analysis.

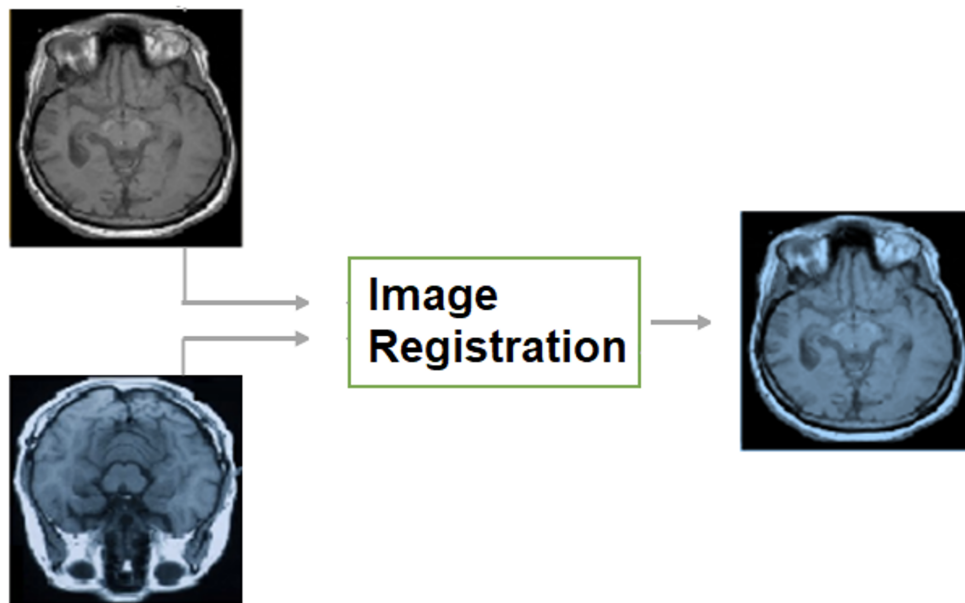


Figure 1.1: General idea of biomedical image registration. A spatial transformation maps the moving image into the coordinate system of the fixed image. This figure is reused from the project proposal.

Registration supports many clinical and research tasks. In longitudinal studies, it allows scans from different dates to be compared voxel by voxel. This is useful for tracking tumor growth, brain atrophy, and treatment response. In multimodal fusion, registration combines complementary information from CT, MRI, PET, or ultrasound. In atlas-based analysis, labels from a reference anatomy can be transferred to a new image. Registration is also used in image-guided surgery, radiotherapy, population studies, and motion correction [1, 3, 4].

The problem has been studied for several decades. Classical techniques solve an optimization problem separately for every image pair. Newer learning-based methods train a neural network once and then predict a transformation in a single forward pass. Classical and learned methods therefore have different strengths. Classical methods are often easier to interpret and do not require training data, while learned methods can be much faster at inference. A useful study should not only compare final accuracy, but should also examine runtime, deformation regularity, and the kind of transformation produced by each method.

1.2 Problem Statement

This project addresses three-dimensional, mono-modal, inter-subject registration of T1-weighted brain MRI. Given a fixed volume F and a moving volume M , the goal is to estimate a transformation ϕ such that the warped image $M \circ \phi$ matches F . Four methods are implemented under a common input, output, and evaluation interface:

1. SimpleITK Diffeomorphic Demons as a classical dense registration method;
2. PSO implemented from scratch to optimize a global 3D affine transform;
3. a compact VoxelMorph-style 3D CNN that predicts a dense field;
4. a compact TransMorph-style 3D Transformer that predicts a dense field.

The project focuses on a functional and reproducible comparison. It does not aim to reproduce every architectural detail, training schedule, and preprocessing step from the original VoxelMorph and TransMorph papers. This distinction is important because the implemented models are deliberately small enough to train and run on one workstation GPU.

1.3 Objectives

The project has the following objectives:

- explain the main mathematical components of image registration using simple and consistent notation;
- implement four standalone registration methods from different method families;
- create synthetic data and an OASIS-1 preprocessing pipeline so the methods can be tested without hidden dependencies;

- train VoxelMorph and TransMorph only on the training split and report final results only on an independent test split;
- compare intensity alignment, anatomical overlap, runtime, success rate, and dense-field regularity;
- provide quantitative plots and qualitative visualizations that make the differences between methods easier to understand.

1.4 Main Contributions

The main contribution is a clean Python repository that places four method families in one reproducible framework. Each method can run independently and saves a registered image, transform or deformation field, overlay image, metrics, and a JSON log. The benchmark script creates a fixed split, trains only on the training pairs, performs learned-model checks on validation pairs, and writes one test-only result table.

The project also adds label-aware evaluation using FreeSurfer segmentation volumes. MRI data are warped using linear interpolation, while segmentation labels are warped using nearest-neighbor interpolation. This allows intensity metrics and anatomical Dice scores to be compared. For dense fields, the benchmark also calculates the percentage of voxels with a non-positive Jacobian determinant as a simple measure of folding.

1.5 Report Organization

Chapter 2 introduces the mathematical background and related work. Chapter 3 describes the OASIS-1 data and preprocessing pipeline. Chapter 4 explains the four implemented methods. Chapter 5 presents the experimental setup and evaluation metrics. Chapter 6 reports quantitative and qualitative results. Chapter 7 discusses trade-offs and limitations. Chapter 8 concludes the report and outlines future work.

The report is organized so that the reader first sees the general registration problem, then the exact implementation choices used in the benchmark, and finally the evidence supporting the comparison. This order is important because the same terms, such as transformation, similarity, Dice, and Jacobian determinant, appear in several later chapters. Introducing them before the experiment makes the result tables easier to interpret and reduces ambiguity when the four methods produce different types of outputs.

Chapter 2

Background and Related Work

2.1 Registration Problem Definition

Let $\Omega \subset \mathbb{R}^d$ denote an image domain, where $d = 2$ for a planar image and $d = 3$ for a volume. Let

$$F : \Omega \rightarrow \mathbb{R}, \quad M : \Omega \rightarrow \mathbb{R} \quad (2.1)$$

be the fixed and moving images. Registration estimates a transformation

$$\phi : \Omega \rightarrow \Omega \quad (2.2)$$

that maps each point x in fixed-image coordinates to a sampling point $\phi(x)$ in the moving image. The registered image is

$$M_{\text{reg}}(x) = (M \circ \phi)(x) = M(\phi(x)). \quad (2.3)$$

The sampling position $\phi(x)$ is usually not located exactly on the discrete voxel grid. Interpolation is therefore required. Linear or trilinear interpolation is suitable for continuous MRI intensity data. Nearest-neighbor interpolation is used for segmentation labels because interpolating label numbers would create invalid anatomical classes.

2.2 Optimization Formulation

A common formulation writes registration as

$$\phi^* = \arg \min_{\phi \in \mathcal{T}} [\mathcal{D}(F, M \circ \phi) + \lambda \mathcal{R}(\phi)], \quad (2.4)$$

where $\mathcal{T}$ is the allowed transformation space, $\mathcal{D}$ measures image mismatch, $\mathcal{R}$ penalizes unrealistic transformations, and λ controls the balance between matching and regularity [5–7].

Equation 2.4 explains why registration can be difficult. A highly flexible transformation can reduce image error but may bend or fold anatomy in an unrealistic way. A very strong regularizer may preserve a smooth shape but fail to align real anatomical differences. The best balance depends on the image modality, anatomy, and application.

2.3 Transformation Models

2.3.1 Rigid Transformation

A 3D rigid transformation contains three rotations and three translations. It preserves distances, angles, and object shape. Rigid registration is appropriate when the main difference is patient or scanner position, but it cannot represent scale differences or local tissue deformation.

2.3.2 Affine Transformation

An affine transform extends rigid motion with scaling and shearing. In homogeneous coordinates it can be written as

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & t_x \\ a_{21} & a_{22} & a_{23} & t_y \\ a_{31} & a_{32} & a_{33} & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}. \quad (2.5)$$

Affine registration models global shape differences while remaining compact and easy to interpret. However, one affine matrix cannot model different local motion in separate brain regions.

2.3.3 Dense Deformable Transformation

A dense transformation assigns a displacement vector $u(x)$ to every voxel:

$$\phi(x) = x + u(x). \quad (2.6)$$

This is much more flexible than a global affine model. It can represent local anatomical differences, but it also requires smoothness control because nearby vectors can otherwise change abruptly.

2.3.4 Diffeomorphic Transformation

A diffeomorphic transformation is smooth and invertible, with a smooth inverse. It is attractive in medical applications because it is designed to preserve topology. Diffeomorphic Demons and SyN are important examples of this family [6, 8]. In practice, numerical approximations can still be checked using the Jacobian determinant.

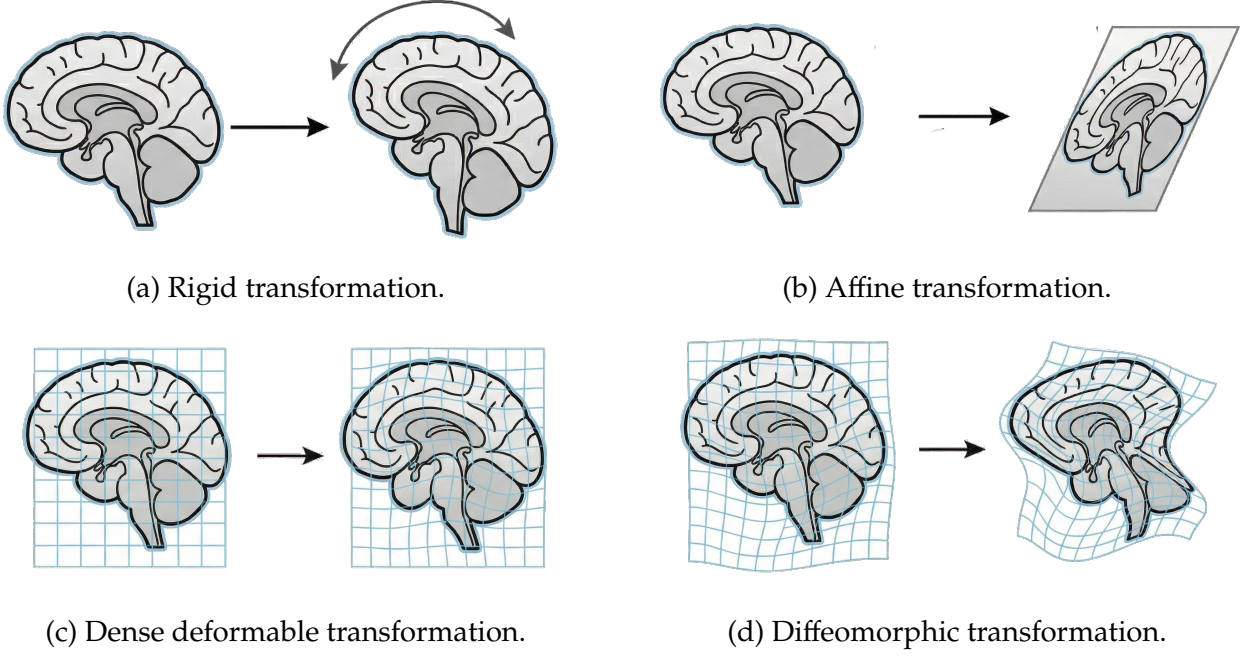


Figure 2.1: Common transformation models in medical image registration. The source illustrations are reused from the project proposal.

2.4 Image Similarity Measures

2.4.1 Mean Squared Error

For images with N voxels, mean squared error is

$$\text{MSE}(F, R) = \frac{1}{N} \sum_{i=1}^N (F_i - R_i)^2, \quad (2.7)$$

where R is the registered image. Lower MSE indicates closer voxel intensities. MSE is simple and differentiable, but it is sensitive to intensity changes and is most suitable for images from the same modality.

2.4.2 Normalized Cross-Correlation

Normalized cross-correlation measures agreement between intensity patterns:

$$\text{NCC}(F, R) = \frac{\sum_i (F_i - \bar{F})(R_i - \bar{R})}{\sqrt{\sum_i (F_i - \bar{F})^2} \sqrt{\sum_i (R_i - \bar{R})^2}}. \quad (2.8)$$

NCC is close to one when the images have strongly correlated patterns. It is less sensitive than MSE to a global linear intensity change and is widely used for mono-modal MRI registration.

2.4.3 Dice Score

Intensity agreement does not directly measure anatomical overlap. If A_l and B_l are the voxel sets for anatomical label l , Dice is

$$\text{Dice}_l(A, B) = \frac{2|A_l \cap B_l|}{|A_l| + |B_l|}. \quad (2.9)$$

In this project, A_l is the set of voxels assigned to label l in the fixed segmentation, while B_l is the set of voxels assigned to the same label after the moving segmentation has been warped into the fixed image space. The numerator $2|A_l \cap B_l|$ counts the common region twice, so the score rewards the part of the anatomy that both segmentations agree on. The denominator $|A_l| + |B_l|$ normalizes this overlap by the total size of the two label regions, which prevents large structures from being judged only by their raw intersection size. Dice lies between 0 and 1: a value of 1 means perfect overlap, a value of 0 means no shared voxels, and intermediate values describe partial anatomical agreement. Because it is computed from label maps rather than MRI intensities, Dice evaluates whether a registration aligns anatomical structures, not merely whether image brightness patterns look similar. This is why Dice can disagree with MSE or NCC: a method may match global intensities well but still place a small structure boundary in the wrong location. Dice is calculated only for evaluation in this report; the learned models are trained without segmentation supervision.

2.5 Regularization and Geometric Validity

The learned dense methods use first-order smoothness regularization. For a three-dimensional field u , a discrete form is

$$\mathcal{R}_{\text{smooth}}(u) = \frac{1}{|\Omega|} \sum_{x \in \Omega} (\|\nabla_x u(x)\|_2^2 + \|\nabla_y u(x)\|_2^2 + \|\nabla_z u(x)\|_2^2). \quad (2.10)$$

This encourages neighboring voxels to move in similar directions.

The local volume change of a transformation is described by

$$J_\phi(x) = \det \left(\frac{\partial \phi(x)}{\partial x} \right). \quad (2.11)$$

A non-positive determinant indicates a local fold or orientation reversal. The benchmark reports the percentage of voxels with $J_\phi(x) \leq 0$. This measure is applicable to the three dense-field methods. It is not reported for PSO because PSO returns one global affine matrix rather than a dense nonlinear field.

2.6 Method Families and Related Work

Classical registration directly optimizes a transformation for each pair. Demons methods use image differences and gradients to build repeated displacement updates. Vercauteren et al. reformulated the method in a diffeomorphic framework to improve transformation composition and topology behavior [8].

Metaheuristic methods treat transformation parameters as a search space. PSO maintains a population of candidate transformations and moves them using personal and global experience. It does not require derivatives and can search a wider parameter region than a local gradient method. PSO has previously been applied to medical registration, including multimodal 3D problems [9–11].

VoxelMorph learns a function that maps an image pair directly to a deformation field. It introduced a practical unsupervised training framework based on an image similarity term, a smoothness term, and a differentiable spatial transformer [12]. TransMorph replaces or extends convolutional feature processing with Transformer blocks to capture longer-range spatial relations [13]. The implementations in this project follow these central ideas, but use smaller networks than the original papers.

Chapter 3

Dataset and Preprocessing

3.1 OASIS-1 Dataset

OASIS-1 is a public cross-sectional brain MRI dataset containing T1-weighted scans from young, middle-aged, and older adults, including nondemented and demented participants [14]. The project uses the FreeSurfer distribution of OASIS-1 rather than the raw scanner files. The FreeSurfer folders provide three-dimensional MRI volumes and anatomical segmentation volumes in MGZ format.

Two files are used from each available scan:

- `mri/T1.mgz`: the T1-weighted brain MRI volume;
- `mri/aparc+aseg.mgz`: the anatomical segmentation used for Dice evaluation.

The benchmark found 425 usable subject scans with both MRI and segmentation information.

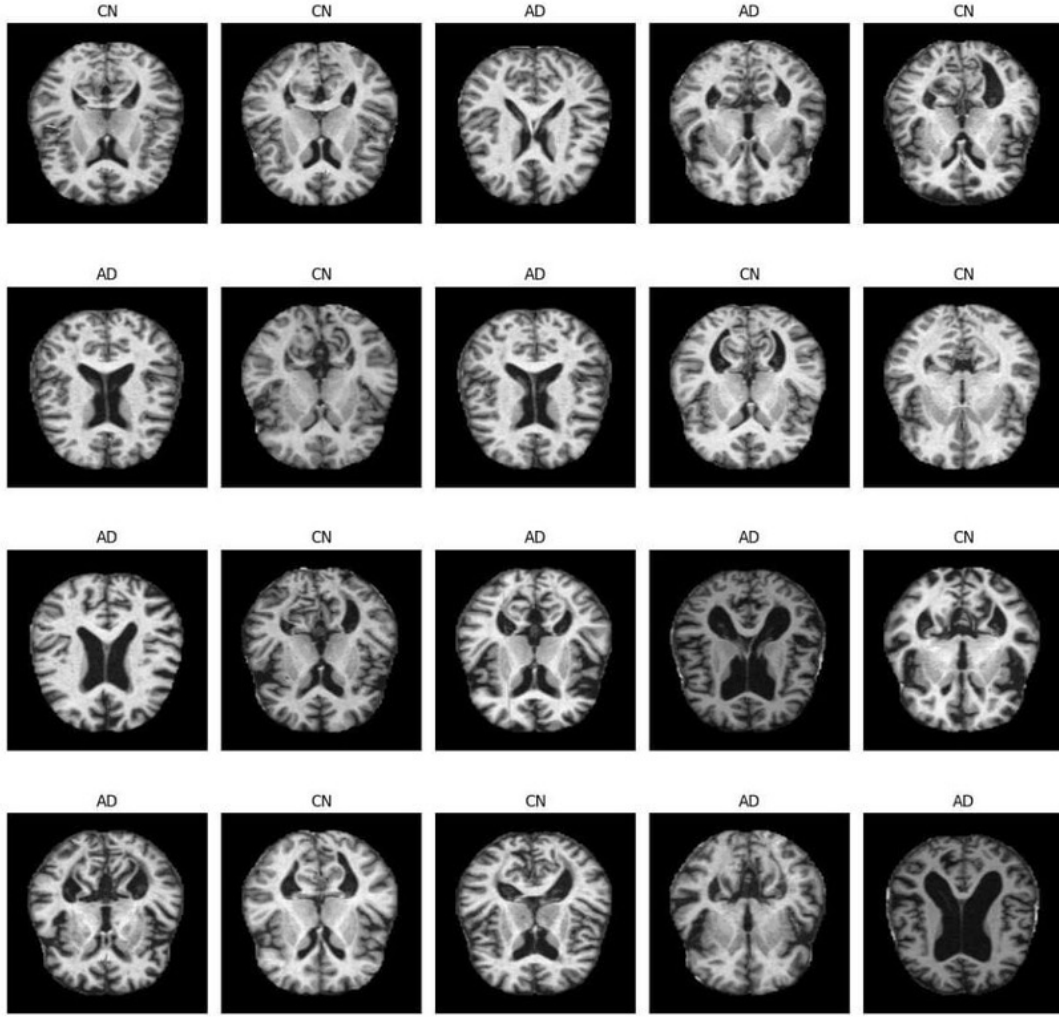


Figure 3.1: Example fixed and moving images from a prepared OASIS-1 pair. The displayed views are taken from the resized 64^3 volumes.

3.2 Preprocessing Pipeline

The preprocessing code is implemented in `scripts/prepare_freesurfer_3d.py`. Each volume is processed as follows:

1. The MGZ volume is loaded with NiBabel and converted to 32-bit floating point values.
2. NaN and infinite values are replaced by zero.
3. For MRI data, nonzero voxels are used to calculate the 1st and 99th intensity percentiles.
4. Intensities are clipped to this percentile range and scaled to $[0, 1]$. This reduces the influence of isolated extreme values.
5. MRI volumes are resized to $64 \times 64 \times 64$ using first-order linear interpolation.
6. Segmentation volumes are resized to the same shape using nearest-neighbor interpolation and rounded to integer label values.

7. The prepared arrays are saved as compressed NIfTI files together with a pair metadata JSON file and a preview image.

The prepared filenames follow a simple convention:

```
fixed_000.nii.gz  
moving_000.nii.gz  
fixed_label_000.nii.gz  
moving_label_000.nii.gz  
pair_000.json  
preview_000.png
```

The deep-learning data loader performs one additional min–max normalization when an image is read. This guarantees a consistent $[0, 1]$ range even if a prepared file contains a small numerical change after interpolation or serialization.

3.3 Pair Construction

Subject folders are sorted by path. Pair i uses subject i as the fixed image and subject $i + 1$ as the moving image. Therefore 425 scans produce 424 adjacent pairs:

$$(S_0, S_1), (S_1, S_2), \dots, (S_{423}, S_{424}). \quad (3.1)$$

This construction is deterministic and provides many inter-subject pairs without random pair generation.

The direction of each pair is kept fixed throughout preprocessing, training, and evaluation. In other words, the first subject in the pair always defines the fixed coordinate space, and the second subject is always warped into that space. The corresponding segmentation files follow the same ordering, so Dice evaluation is performed between the fixed label map and the warped moving label map.

This adjacent-pair strategy is simple and reproducible, but it is not the only possible pairing scheme. Random all-to-all pairing would create many more pairs, while atlas-to-subject pairing would compare every subject against a common reference. The adjacent design is chosen here because it keeps the benchmark size manageable and avoids introducing an additional atlas selection step.

3.4 Train, Validation, and Test Split

The benchmark uses seed 123 to randomly permute the 424 pair indices. The split is created before learned-model training and is saved to `splits.json`.

Table 3.1: Dataset split used in the final benchmark.

Split	Number of pairs	Percentage
Training	297	70%
Validation	42	10%
Test	85	20%
Total	424	100%

VoxelMorph and TransMorph are trained only on the 297 training pairs. Learned methods are checked on validation pairs. All final quantitative results in this report use only the 85 test pairs. Classical Demons and PSO do not require training and are run directly on the same test pairs.

3.5 Anatomical Labels

Dice is averaged over 20 selected FreeSurfer labels, covering left and right white matter, cerebral cortex, lateral ventricles, thalamus, caudate, putamen, pallidum, hippocampus, amygdala, and accumbens area. The source is the volumetric `aparc+aseg.mgz` or `aseg.mgz` segmentation. Surface `label/* .label` files are not used.

The moving segmentation is transformed using the same estimated transformation as the moving MRI, but with nearest-neighbor sampling. This preserves discrete label identities. The fixed segmentation is never used as an input to the registration method; it is used only for evaluation.

3.6 Dataset Limitations

The preprocessing is intentionally lightweight, but several limitations should be stated clearly.

First, the split is performed at the pair level. Adjacent pairs share subjects, for example (S_0, S_1) and (S_1, S_2) . If these pairs are assigned to different splits, the same subject can appear in both training and test data. This is a form of subject leakage. A stronger future protocol should split unique subjects first and create pairs only within each split.

Second, direct resizing to 64^3 reduces fine anatomical detail. The script also does not preserve the complete original physical spacing and orientation when it writes the prepared arrays. Third, the project does not apply additional N4 bias field correction, atlas pre-alignment, or augmentation. The benchmark should therefore be interpreted as a controlled functional comparison, not as a paper-level clinical validation.

Chapter 4

Implemented Registration Methods

4.1 Shared Processing Pipeline

All four methods receive the same fixed and moving NIfTI volumes. Image values are normalized, the method estimates a transform, and the moving image is resampled in the fixed coordinate system. Every run writes a registered image, an overlay, metrics, and a JSON log. Dense methods additionally save a displacement field in NumPy and NIfTI formats. PSO saves affine parameters and a 4×4 matrix.

VoxelMorph and TransMorph share the same spatial transformer and loss. If the network predicts a displacement field u , the sampling transformation is

$$\phi(x) = x + u(x), \quad M_{\text{reg}}(x) = M(\phi(x)). \quad (4.1)$$

PyTorch `grid_sample` performs trilinear sampling during training and inference. Coordinates are normalized to the interval $[-1, 1]$ as required by this operation.

4.2 Classical Diffeomorphic Demons

4.2.1 Method Idea

Demons registration repeatedly estimates small displacement updates that reduce the difference between fixed and moving images. The update is smoothed to avoid abrupt local motion and then combined with the current transformation. The diffeomorphic formulation uses transformation composition to better preserve a smooth and invertible mapping [8].

The method can be understood as a classical iterative alternative to the learned dense-field models. Instead of predicting the entire deformation with a neural network, Demons updates the transformation step by step from the current image mismatch. Regions with strong and reliable image gradients provide more useful motion information, while nearly homogeneous regions are harder to align because many local displacements can produce similar intensity values.

The role of smoothing is especially important in medical registration. Without regularization, neighboring voxels could move in inconsistent directions and the field

would become anatomically implausible. Gaussian smoothing spreads local updates over nearby voxels, encouraging the deformation to vary gradually across the volume.

The conceptual procedure is summarized as pseudocode in Algorithm 1.

Input: Fixed image F , moving image M , number of iterations T

Output: Dense displacement field u and registered image

Initialize the displacement field $u_0(x) = 0$;

for $t \leftarrow 1$ **to** T **do**

warp the moving image using the current field u_{t-1} ;
 compute an intensity-driven update field Δu_t ;
 smooth Δu_t with a Gaussian regularizer;
 compose the smoothed update with the current transformation;
 store the composed field as u_t ;

end

Warp M with u_T using linear interpolation;

Algorithm 1: Pseudocode for Diffeomorphic Demons registration

4.2.2 Project Implementation

The code uses `SimpleITK.DiffeomorphicDemonsRegistrationFilter` in `src/methods/classical/register.py`. Both images are cast to floating point and rescaled to $[0, 1]$. The final experiment uses 10 iterations and a Gaussian standard deviation of 1.3 for field smoothing. The moving MRI is resampled linearly, while labels are transformed later with nearest-neighbor interpolation.

The implementation treats Demons as a test-time optimizer. For every pair, the algorithm starts from an identity displacement field and iteratively improves the alignment for that pair only. No information is shared across pairs, and no model parameters are saved for later inference. This makes the method useful as a training-free baseline, but it also means the optimization cost must be paid again for every new fixed–moving pair.

The method has no training phase. Its optimization occurs separately for each test pair. It returns a dense field, so displacement magnitude, Jacobian determinant, and folding can be inspected.

4.2.3 Expected Strengths and Weaknesses

Demons is a useful baseline because it is established, deterministic under a fixed configuration, and easy to run without training data. Its explicit smoothing often gives stable fields. However, the result depends on iteration count, smoothing, initialization, and image gradient quality. With only 10 iterations, the baseline is intentionally small and may not reach the strongest possible Demons result.

Another important limitation is that Demons is mainly intensity driven. If two structures have similar intensities or if the local gradient is weak, the update direction may be uncertain. The method also does not learn dataset-level anatomy: it cannot use experience from previous registrations to guide a difficult new pair. These properties explain why Demons is stable in this benchmark but does not dominate all accuracy metrics.

4.3 Particle Swarm Optimization for 3D Affine Registration

4.3.1 Affine Parameterization

The PSO method is implemented from scratch in `src/methods/metaheuristic/ps.py`. Each particle represents a 12-dimensional 3D affine transform:

$$\theta = (r_x, r_y, r_z, t_x, t_y, t_z, s_x, s_y, s_z, h_{xy}, h_{xz}, h_{yz}), \quad (4.2)$$

where r denotes rotation, t translation, s scale, and h shear. The rotation range is $[-8^\circ, 8^\circ]$. Translation is bounded according to the volume size, scales are in $[0.90, 1.10]$, and shears are in $[-0.08, 0.08]$.

This parameterization deliberately restricts PSO to global alignment. All voxels are transformed by the same affine matrix, so the method can correct head pose, overall scale, and simple shape skew, but it cannot bend one anatomical region independently of another. This makes the search space much smaller than a dense deformable field and keeps the affine result easy to inspect.

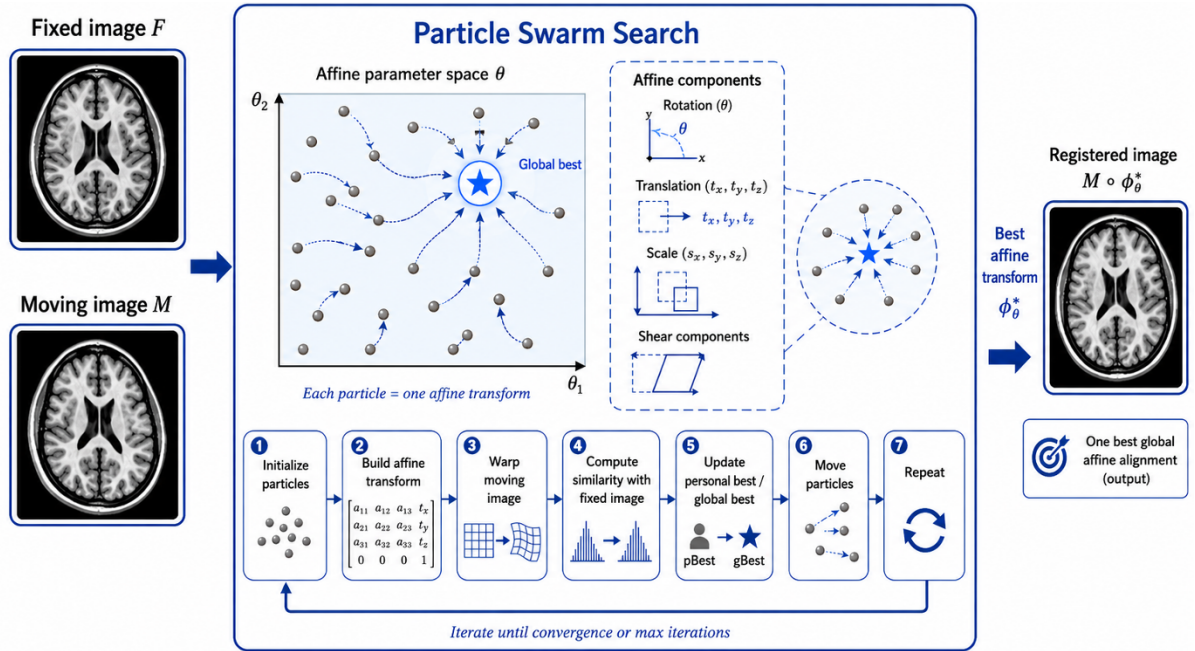


Figure 4.1: Conceptual workflow of PSO affine registration. Each particle is one candidate affine transformation. The figure was extracted from the project presentation.

4.3.2 Particle Update

For particle i at iteration t , velocity and position are updated by

$$v_i^{t+1} = \omega v_i^t + c_1 r_1 (p_i - x_i^t) + c_2 r_2 (g - x_i^t), \quad (4.3)$$

$$x_i^{t+1} = \text{clip}(x_i^t + v_i^{t+1}, \theta_{\min}, \theta_{\max}), \quad (4.4)$$

where p_i is the personal best, g is the global best, and r_1, r_2 are random vectors in $[0, 1]$. The implementation uses inertia $\omega = 0.72$ and $c_1 = c_2 = 1.45$. One particle is initialized as the identity transform so that PSO cannot start worse than the unregistered image.

For each candidate, the moving image is affinely sampled and scored. The final benchmark uses NCC, so the minimization objective is

$$\mathcal{D}_{\text{PSO}} = 1 - \text{NCC}(F, M \circ \phi_{\theta}). \quad (4.5)$$

The final run uses 16 particles and 30 iterations. PSO therefore evaluates 480 candidate states per pair, which explains its higher runtime.

The personal-best term encourages each particle to return toward the best solution it has found so far, while the global-best term pulls the whole swarm toward the best solution found by any particle. The random multipliers prevent all particles from following exactly the same trajectory. Clipping is used after each update so that rotations, translations, scales, and shears remain inside the predefined medically reasonable search range.

4.3.3 Expected Strengths and Weaknesses

PSO does not require gradients and can explore rotations, translations, scales, and shears jointly. A global affine transform is also easy to inspect and cannot produce local dense-field folding. However, PSO is computationally expensive because every particle requires a full volume warp and score. It also cannot model local anatomical deformation because one affine matrix controls the whole brain.

For this reason, PSO is best viewed as a global registration method rather than a complete deformable solution. It can be valuable when the dominant mismatch is pose or scale, and it can serve as a pre-alignment stage before a dense method. When the remaining error is local anatomical variation, however, a dense field from Demons, VoxelMorph, or TransMorph is more expressive.

4.4 Compact VoxelMorph-Style CNN

4.4.1 Reference Idea

VoxelMorph is a representative example of learning-based deformable registration. Instead of optimizing a new deformation field independently for every test image pair, it trains a neural network once and then reuses the learned parameters for new pairs. In this sense, VoxelMorph performs *amortized registration*: the costly iterative optimization is shifted from test time to training time.

For a fixed image F and moving image M , VoxelMorph learns a function g_{θ} that predicts a dense displacement field:

$$u = g_{\theta}(F, M). \quad (4.6)$$

The transformation is then written as

$$\phi(x) = x + u(x), \quad (4.7)$$

where x is a voxel coordinate and $u(x)$ is the predicted displacement at that coordinate. The moving image is sampled at the transformed coordinates using a differentiable spatial transformer, producing $M \circ \phi$. Because the warping operation is differentiable, gradients

from the image-similarity loss can flow through the sampler back into the CNN. This is the main reason the model can be trained without ground-truth deformation fields [12].

The original VoxelMorph framework uses an encoder–decoder convolutional network similar to a U-Net. The encoder progressively extracts contextual features from the concatenated image pair, while the decoder upsamples these features back to the image resolution. Skip connections preserve fine spatial detail that would be lost if the decoder relied only on the bottleneck representation. The final registration layer predicts one displacement component for each spatial axis.

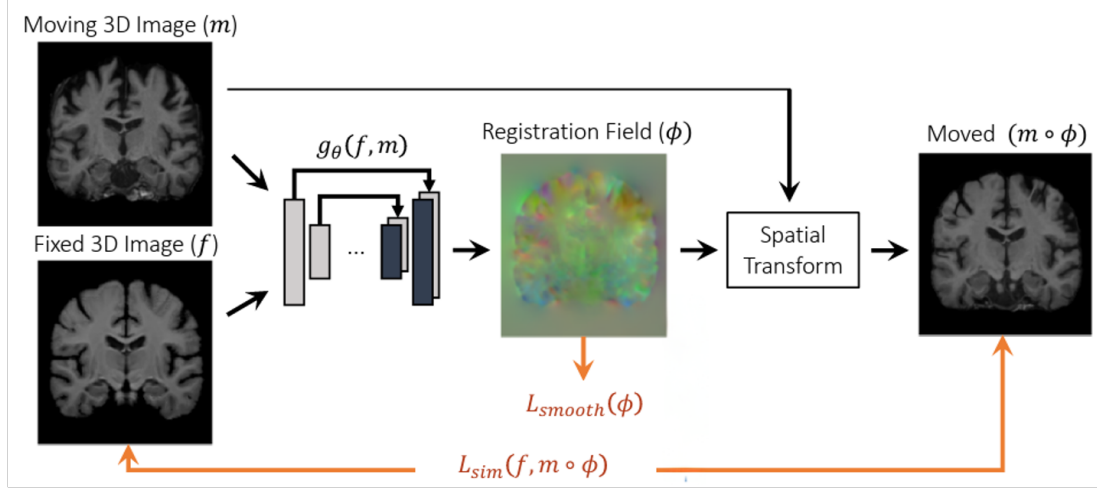


Figure 4.2: General VoxelMorph learning framework. The project implementation follows the same input–field–warp–loss idea but uses a smaller 3D CNN. The figure was extracted from the project presentation and is based on Balakrishnan et al. [12].

4.4.2 Project Architecture

The project follows the VoxelMorph input–field–warp–loss pipeline, but the CNN is intentionally much smaller than the network used in the original paper. Fixed and moving images are concatenated channel-wise to form an input tensor of shape $B \times 2 \times 64 \times 64 \times 64$. The compact model contains:

- a first 3D convolution block with four channels;
- average pooling followed by an eight-channel encoder block;
- a second pooling operation and an eight-channel bottleneck;
- trilinear upsampling with skip concatenation;
- two decoder blocks;
- a final 3D convolution that outputs three displacement components.

Each block uses two $3 \times 3 \times 3$ convolutions with Leaky ReLU activation. The skip connections concatenate encoder features with decoder features at matching spatial scales. This preserves local information from high-resolution feature maps while still allowing the bottleneck to represent broader context.

The final flow layer is initialized with very small weights so that the predicted field is close to zero at the start of training. Consequently, the initial transformation is close to the identity mapping. This mirrors the practical registration assumption that the network should learn deformations gradually rather than begin with large arbitrary warps.

4.4.3 Training Objective

The training objective follows the same unsupervised structure as VoxelMorph: an image-matching term encourages the warped moving image to resemble the fixed image, while a regularization term discourages spatially irregular deformation. The project loss is

$$\mathcal{L}_{\text{VM}} = \text{MSE}(F, M \circ \phi) + 0.05\mathcal{R}_{\text{smooth}}(u). \quad (4.8)$$

Here, the MSE term measures voxel-wise intensity disagreement after warping. The smoothness term penalizes neighboring displacement differences and can be written conceptually as

$$\mathcal{R}_{\text{smooth}}(u) = \frac{1}{|\Omega|} \sum_{x \in \Omega} \sum_{d \in \{x, y, z\}} \|\nabla_d u(x)\|_2^2, \quad (4.9)$$

where Ω is the voxel domain. This term does not make the deformation strictly diffeomorphic, but it biases neighboring voxels to move coherently.

The model is optimized with Adam at a learning rate of 10^{-3} . It is trained for 20 epochs with batch size 2. The checkpoint with the lowest training loss is saved as `best.pt`. At test time, no pair-specific optimization is performed: registration requires only one CNN forward pass, followed by the same spatial transformer used during training.

4.4.4 Relation to the Original Paper

This implementation should be interpreted as a compact VoxelMorph-style model, not a full reproduction of the original VoxelMorph experiments. It preserves the central methodological ideas from the paper: paired image input, CNN-based dense field prediction, differentiable warping, and unsupervised optimization through image similarity plus deformation regularization. However, it uses lower channel counts, a smaller input resolution, MSE rather than every similarity option studied in the paper, and a simple displacement field rather than reproducing all probabilistic or diffeomorphic variants discussed in the broader VoxelMorph framework [12].

4.4.5 Expected Strengths and Weaknesses

After training, VoxelMorph predicts a field in one forward pass, which makes it much faster than repeated pair-wise optimization. Convolutional kernels also give the model a useful locality prior: nearby voxels are processed together, which is well matched to smooth anatomical deformation. The encoder–decoder structure combines local detail with coarser context, and the dense output can model local non-rigid motion that a global affine method cannot represent.

The main limitations follow directly from these design choices. The learned model depends on the training distribution and may generalize poorly to images with different preprocessing, anatomy, or scanner characteristics. Smoothness is only a soft penalty, so

the output field can still contain folding. Finally, because the project CNN is deliberately compact, its effective receptive field and feature capacity are limited compared with larger VoxelMorph configurations.

4.5 Compact TransMorph-Style Transformer

4.5.1 Reference Idea

TransMorph extends the learning-based registration idea by replacing a purely convolutional feature extractor with Transformer-based feature processing. The motivation is that deformable registration may require correspondences between spatially distant anatomical regions. A small CNN can model such relationships only indirectly by stacking many local operations, whereas attention allows token features to interact more explicitly.

For query, key, and value matrices Q , K , and V , scaled dot-product attention is

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V, \quad (4.10)$$

where d_k is the key dimension. Multi-head attention repeats this operation in several learned subspaces and combines the results. In registration, this helps the network encode broader anatomical context before predicting a dense field.

The original TransMorph architecture is a hybrid Transformer–ConvNet framework for unsupervised volumetric registration. It uses a Swin Transformer-style encoder to build hierarchical contextual features and a convolutional decoder to recover a dense deformation field at full image resolution. The warped image is then computed with a spatial transformer, and the model is trained with image similarity and deformation regularization terms, as in other unsupervised registration networks [13].

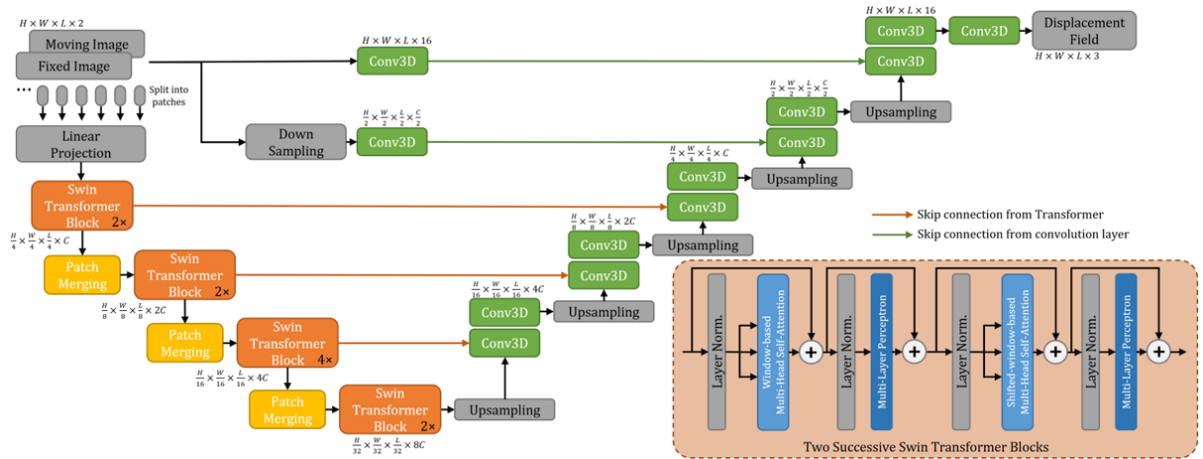


Figure 4.3: Reference TransMorph architecture. The project uses a compact Transformer encoder and convolutional decoder rather than the complete Swin Transformer architecture shown here. The figure was extracted from the project presentation and is based on Chen et al. [13].

4.5.2 Project Architecture

The project keeps the high-level TransMorph idea but uses a much smaller Transformer. Fixed and moving volumes are concatenated into a two-channel input. A 3D convolution with kernel size 8 and stride 8 partitions the 64^3 volume into an $8 \times 8 \times 8$ grid of non-overlapping patch tokens. Thus, each image pair is represented by 512 tokens. Each token has embedding dimension 24.

A sinusoidal 3D positional encoding is added so that the Transformer receives information about where each patch came from in the volume. The encoded tokens are passed through one Transformer encoder layer with four attention heads. The feed-forward dimension is four times the embedding size. Compared with the original TransMorph, this removes the deep hierarchical Swin Transformer stages and shifted-window attention, but it retains the essential tokenization and self-attention mechanism.

The encoded sequence is reshaped into a 3D feature grid. Three convolutional decoder stages progressively upsample the grid by a factor of two until the original image resolution is reached. GELU activations are used in the decoder. A final 3D convolution outputs the three-component dense displacement field. This decoder is important because registration is a dense prediction problem: even if attention operates on coarse patch tokens, the final output must assign a displacement vector to every voxel.

The training objective, spatial transformer, optimizer, learning rate, number of epochs, and smoothness weight are the same as for VoxelMorph. This keeps the comparison focused on the feature extractor rather than a different loss.

4.5.3 Relation to the Original Paper

The implementation is therefore best described as TransMorph-style rather than a complete TransMorph reproduction. It follows the paper’s main design principle: combine Transformer context modeling with a convolutional decoder for dense medical image registration. However, the original TransMorph uses a richer hierarchical Swin Transformer encoder, stronger multi-scale feature construction, paper-specific hyperparameters, and additional variants such as diffeomorphic and Bayesian formulations. The project model deliberately omits these components to fit the computational constraints of a small comparative benchmark [13].

4.5.4 Expected Strengths and Weaknesses

Transformer attention can capture broader context more directly than a small CNN, which is useful when local appearance alone is ambiguous. The hybrid design also keeps convolutional decoding, so the method still has a mechanism for recovering voxel-level displacement detail. Like VoxelMorph, inference requires only one network forward pass and one spatial warp.

The compact Transformer also has limitations. Patch embedding reduces spatial resolution at the start, so fine details must be reconstructed by the decoder. Attention layers require more memory than simple convolutions, and a very small Transformer may not fully exploit the advantages reported by the original TransMorph paper. Since the model uses the same soft smoothness loss as VoxelMorph, it can still produce non-diffeomorphic

regions even if its deformation field is empirically smoother in some test cases.

4.6 Comparison of Implemented Methods

Table 4.1: Main characteristics of the four implementations.

Method	Transformation	Training	Test-time process	Main output
Classical	Dense diffeomorphic	No	Iterative optimization for each pair	Dense displacement field
PSO	Global 3D affine	No	Population search for each pair	12 parameters and affine matrix
VoxelMorph	Dense deformable	Yes	One CNN forward pass	Dense displacement field
TransMorph	Dense deformable	Yes	One Transformer forward pass	Dense displacement field

Chapter 5

Experimental Setup

5.1 Benchmark Protocol

The final experiment is the run stored under `outputs/benchmark/oasis1_3d_functional_clean_affine_gpu0_retry_20260601_190859`. It is described as an OASIS-1 FreeSurfer-derived 3D functional benchmark. It is not an exact reproduction of the original VoxelMorph or TransMorph studies.

The two learned models are trained from scratch on the training split. Validation pairs are used only to check learned-model behavior. Final tables use test pairs only. Classical and PSO methods are evaluated directly on the same 85 test pairs. All four methods complete all 85 pairs, so the final comparison has no failed runs.

This protocol separates model selection from final reporting. Training pairs are used to update VoxelMorph and TransMorph weights, validation pairs are used as a sanity check during development, and test pairs are held out for the final comparison. Classical Demons and PSO do not learn from the training split, but they are still evaluated on the identical test pairs so that the metric averages are computed over the same fixed-moving cases.

The benchmark records both image outputs and numerical summaries. For each test pair, the registered image is saved along with the deformation or affine transform, overlay visualization, runtime, intensity metrics, Dice scores, and Jacobian statistics when applicable. This makes the final tables traceable to individual pair-level outputs rather than only to aggregate numbers.

5.2 Hardware and Software

Table 5.1: Execution environment reported in the project presentation.

Component	Configuration
CPU	Intel Xeon Gold 6244, 16 cores
Memory	64 GB RAM
GPU	NVIDIA Quadro RTX 6000, 24 GB VRAM
Python	3.12.13
PyTorch	2.10.0
CUDA	12.6
Classical library	SimpleITK

CUDA device 0 is exposed to the final clean-affine run and is addressed as `cuda:0`. Learned-model training is sequential. A deterministic seed of 123 is used for data splitting, model initialization, data loading, and PSO.

The reported runtimes should therefore be interpreted as implementation- and hardware-specific measurements. They are useful for comparing the four methods inside this project because the benchmark is run under one environment, but they should not be treated as universal runtime constants. Different GPUs, CPU memory bandwidth, interpolation libraries, or batch-processing strategies could change the absolute values.

5.3 Method Hyperparameters

Table 5.2: Main configuration of the four methods.

Method	Configuration
Classical	Diffeomorphic Demons; 10 iterations; smoothing standard deviation 1.3
PSO	3D affine; 16 particles; 30 iterations; NCC objective; inertia 0.72; cognitive/social weights 1.45
VoxelMorph	3D CNN; base channels 4; 20 epochs; batch size 2; Adam; learning rate 10^{-3} ; MSE loss; smoothness weight 0.05
TransMorph	Embedding 24; patch size 8; one Transformer layer; four heads; decoder channels 12; same training loss and optimizer as VoxelMorph

The hyperparameters are intentionally modest. Classical Demons uses a small iteration count, PSO uses a limited swarm and search budget, and both learned models are compact enough to train on downsampled 64^3 volumes. This design keeps the comparison runnable in a course-project setting. The trade-off is that the benchmark measures these practical implementations rather than the maximum performance that each method family could achieve after extensive tuning.

5.4 Evaluation Metrics

For each test pair, the benchmark first measures MSE and NCC between fixed and unregistered moving images. It then measures the same metrics after registration. The change values are

$$\Delta\text{MSE} = \text{MSE}_{\text{before}} - \text{MSE}_{\text{after}}, \quad (5.1)$$

$$\Delta\text{NCC} = \text{NCC}_{\text{after}} - \text{NCC}_{\text{before}}. \quad (5.2)$$

Positive values indicate improvement.

Dice is computed for each selected anatomical label before and after registration, then averaged over labels available in the pair. Runtime measures the complete method call for one pair, including inference or pair-wise optimization and output generation. The benchmark also records success and failure counts. For dense fields, Jacobian folding percentage is the fraction of voxels where the determinant is non-positive.

5.5 Output and Reproducibility

The final benchmark writes `benchmark_results.csv`, `benchmark_results.json`, `benchmark_summary.md`, `benchmark_summary.json`, `validation_results.csv`, and `splits.json`. Per-pair folders contain registered images, overlays, deformation fields or affine parameters, label warps, and logs. Training folders contain resolved configurations, history logs, and best/last checkpoints.

The main benchmark can be launched using

```
CUDA_VISIBLE_DEVICES=0 \  
bash scripts/run_oasis1_3d_clean_affine_fast.sh
```

The repository also includes smoke-test scripts for synthetic and FreeSurfer data.

Chapter 6

Results

6.1 Overall Quantitative Results

Before registration, the mean test MSE is 0.021801, mean NCC is 0.601383, and mean anatomical Dice is 0.127248. Table 6.1 presents the final after-registration results.

Table 6.1: Mean results on 85 OASIS-1 test pairs. Lower MSE and runtime are better; higher NCC and Dice are better. Folding is reported only for dense fields.

Method	Success	Time (s)	MSE	Δ MSE	NCC	Δ NCC	Dice	Δ Dice	Folding (%)
Classical	85/85	0.5117	0.013728	0.008073	0.732291	0.130908	0.164172	0.036924	0.000000
PSO affine	85/85	9.3619	0.010877	0.010924	0.793471	0.192088	0.330794	0.203545	N/A
VoxelMorph	85/85	0.3631	0.008916	0.012885	0.827261	0.225878	0.164668	0.037419	0.158458
TransMorph	85/85	0.3852	0.009916	0.011885	0.807193	0.205810	0.218955	0.091706	0.034584

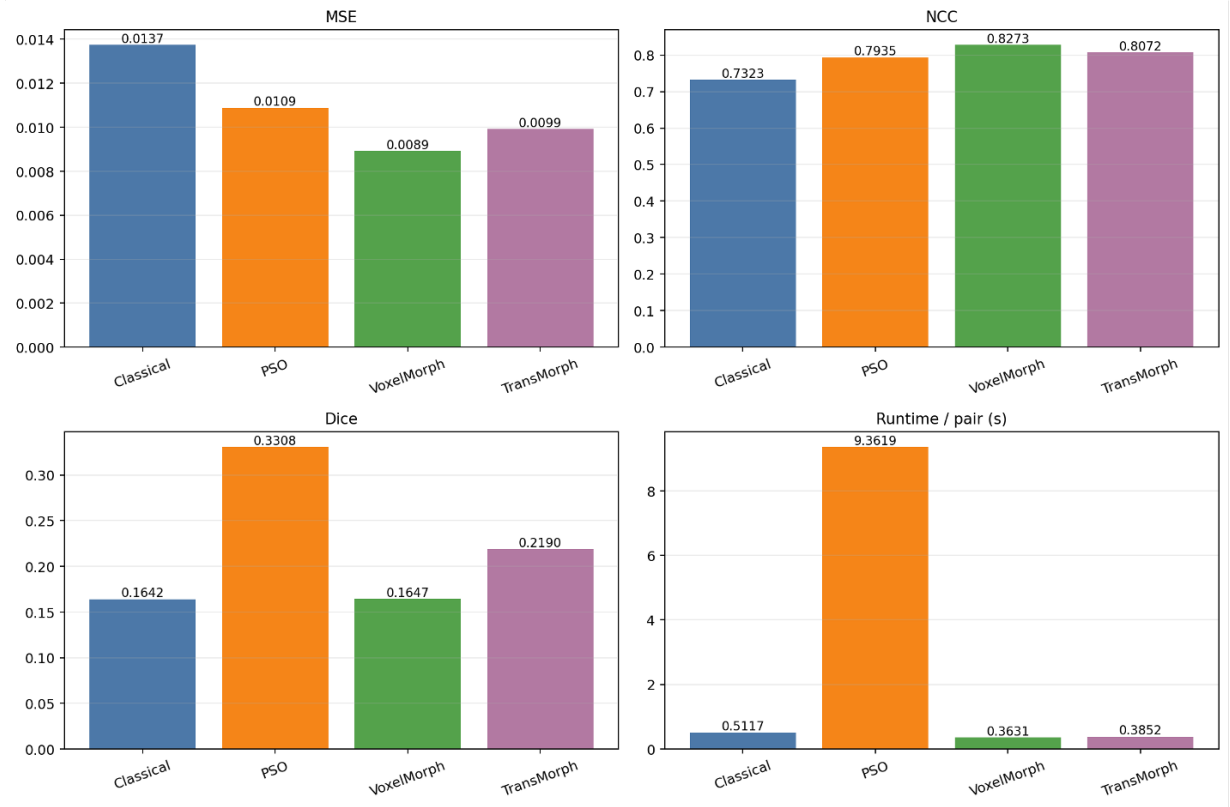


Figure 6.1: Mean test MSE, NCC, Dice, and runtime for the four methods.

VoxelMorph is strongest on intensity alignment. It has the lowest MSE, highest NCC, largest mean improvement in both intensity metrics, and shortest mean runtime. PSO is strongest on anatomical overlap, with Dice 0.330794. Its Dice improvement is more than twice that of TransMorph, but its runtime is roughly 25 times that of the learned models. TransMorph is second among the learned methods for intensity metrics but provides clearly higher Dice and lower folding than VoxelMorph. Classical has weaker intensity and Dice results, but no detected folding.

6.2 Metric Distributions

A mean value can hide difficult cases or unstable behavior. Figure 6.2 shows the distribution across all 85 test pairs.

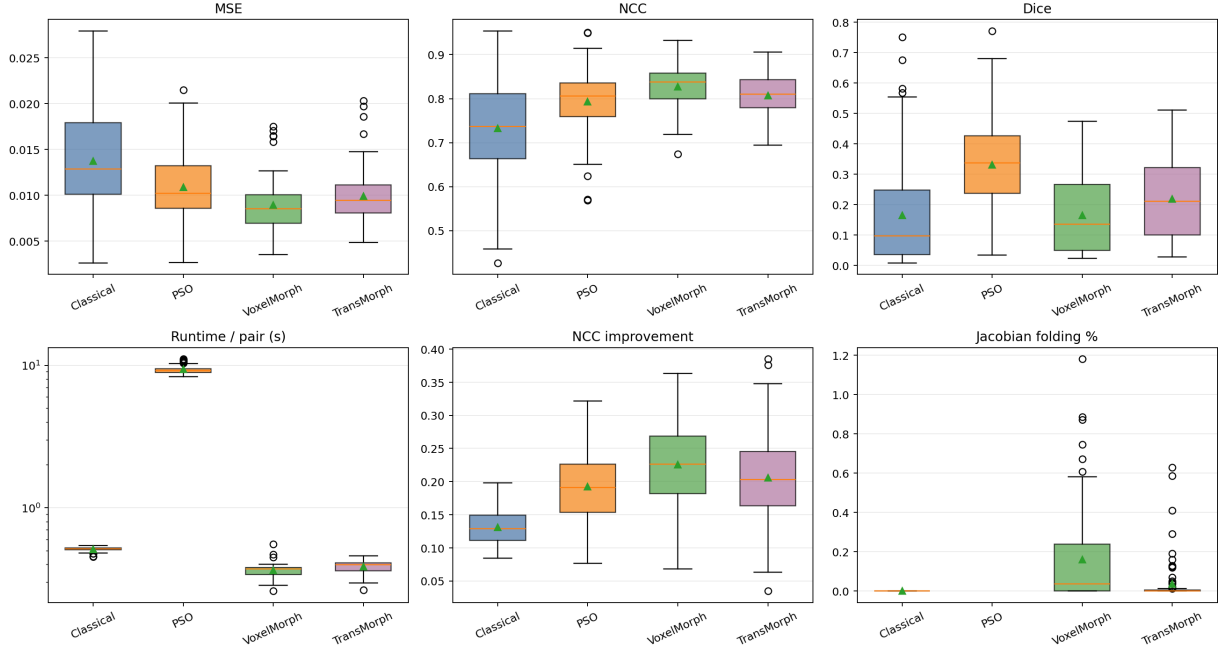


Figure 6.2: Distribution of test metrics across image pairs.

The distributions show that anatomical Dice varies substantially between pairs. Part of this variation is already present before registration because some adjacent subjects have more similar anatomy than others. PSO shifts the Dice distribution upward in many cases, while VoxelMorph gives the strongest and most consistent NCC values. Runtime has a clear separation: PSO is much slower, while VoxelMorph and TransMorph cluster near a fraction of one second.

6.3 Accuracy–Runtime Trade-off

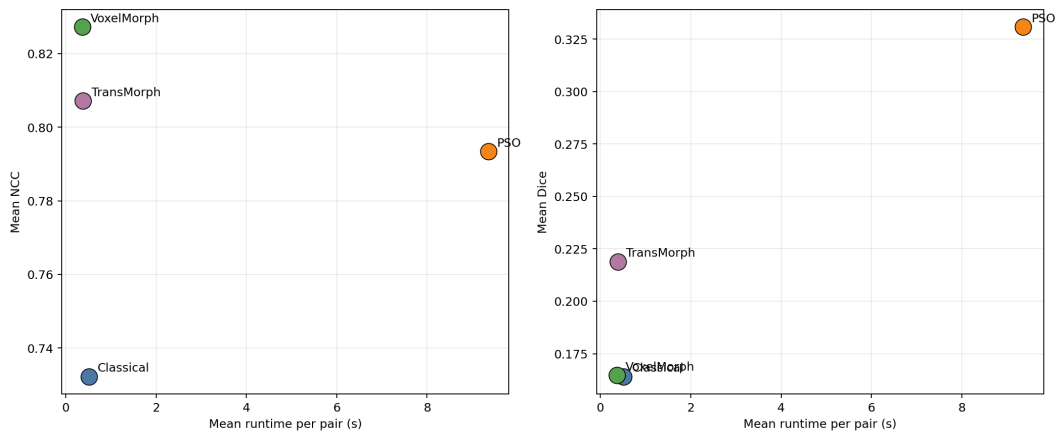


Figure 6.3: Mean runtime compared with NCC and Dice quality.

Figure 6.3 makes the trade-off easy to see. VoxelMorph is the most attractive option when fast intensity alignment is the main goal. PSO occupies a different operating point: it spends much more computation to search the affine parameter space and produces the

best label overlap. TransMorph lies between these methods and offers fast inference with better anatomical overlap than VoxelMorph.

The figure also shows why runtime should be interpreted together with the type of transformation. VoxelMorph and TransMorph perform most of their optimization during training, so test-time registration is only a forward pass through the network and one resampling operation. PSO has no training cost, but it performs a large search for every test pair. Classical Demons also optimizes each pair at test time, but with the small iteration count used here its runtime remains below PSO. Thus, a method that is cheap at inference may have required training, while a training-free method may still be expensive per pair.

6.4 Per-Pair Winners

Table 6.2: Number of test pairs won by each method under selected criteria.

Criterion	Classical	PSO	VoxelMorph	TransMorph
Highest NCC	8	10	54	13
Highest Dice	9	55	6	15
Fastest runtime	0	0	66	19
Lowest folding among dense methods	85	N/A	0	0

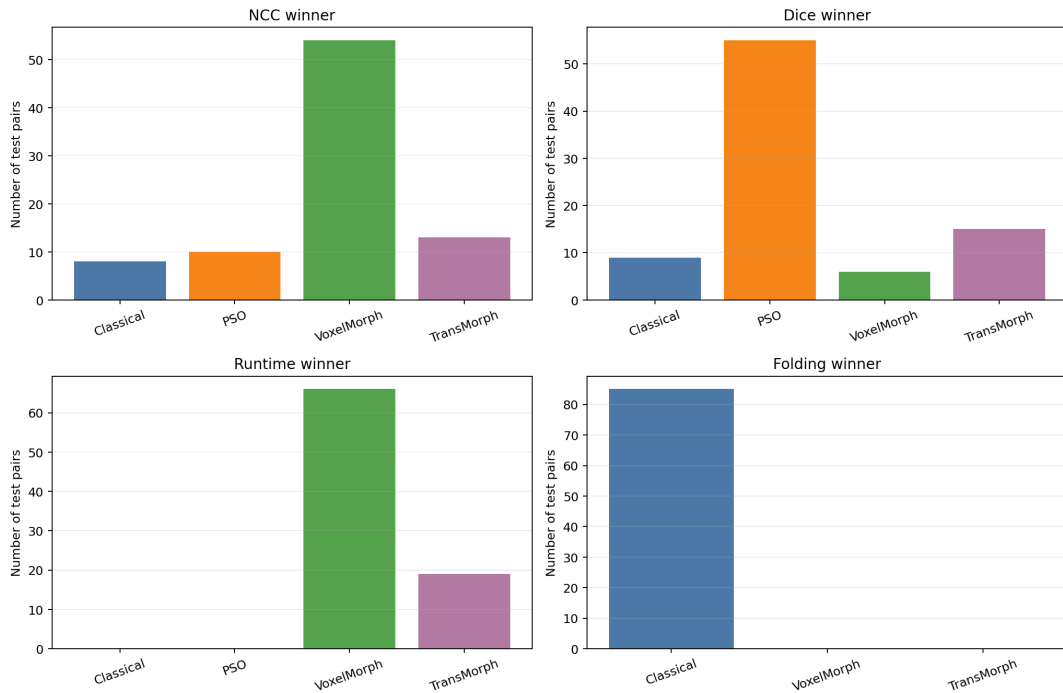


Figure 6.4: Per-pair winner counts for major benchmark criteria.

The win counts confirm that the mean results are not caused by only a small number of pairs. VoxelMorph wins NCC on 54 of 85 pairs, while PSO wins Dice on 55 pairs. TransMorph wins more Dice cases than VoxelMorph, supporting the interpretation that it is the stronger learned model for anatomical overlap in this experiment.

Per-pair winners are useful because averages can hide heterogeneous behavior. For example, a method may have a good mean Dice because it performs extremely well on a subset of easier pairs, even if it is less reliable on harder pairs. Counting winners does not replace the mean and standard deviation, but it provides another view of robustness across the test set. In this benchmark, the winner counts are consistent with the metric averages: the same methods that lead the mean tables also win many individual cases.

6.5 Relationship Between NCC and Dice

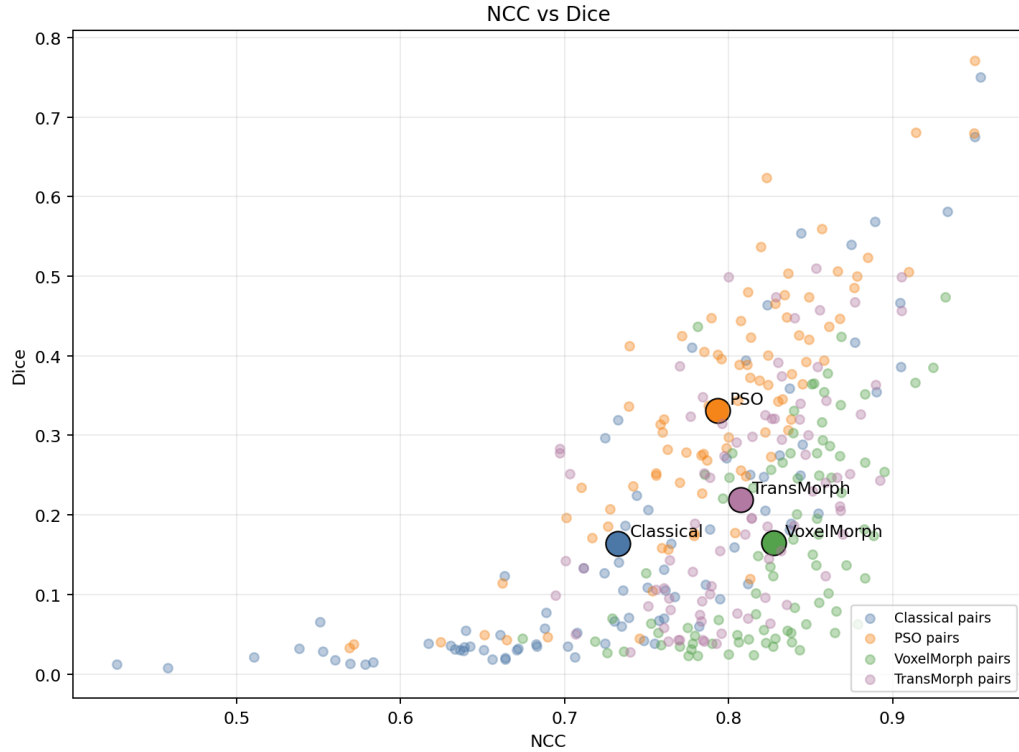


Figure 6.5: Pair-level relationship between after-registration NCC and Dice.

NCC and Dice are related, but Figure 6.5 shows that they are not equivalent. A method can improve the overall intensity pattern while still misaligning the boundary of a small anatomical region. Conversely, a global affine adjustment can improve the overlap of major structures without achieving the smallest voxel-wise intensity error. This explains why VoxelMorph leads NCC while PSO leads Dice.

This difference is especially important in medical imaging because downstream tasks may depend on different aspects of alignment. Intensity-based analysis, such as voxel-wise comparison of T1 signal patterns, benefits from high NCC and low MSE. Label transfer or atlas-based segmentation, however, depends more directly on anatomical boundary overlap. A registration method should therefore be selected according to the downstream task rather than according to one metric alone.

6.6 Anatomical Label Results

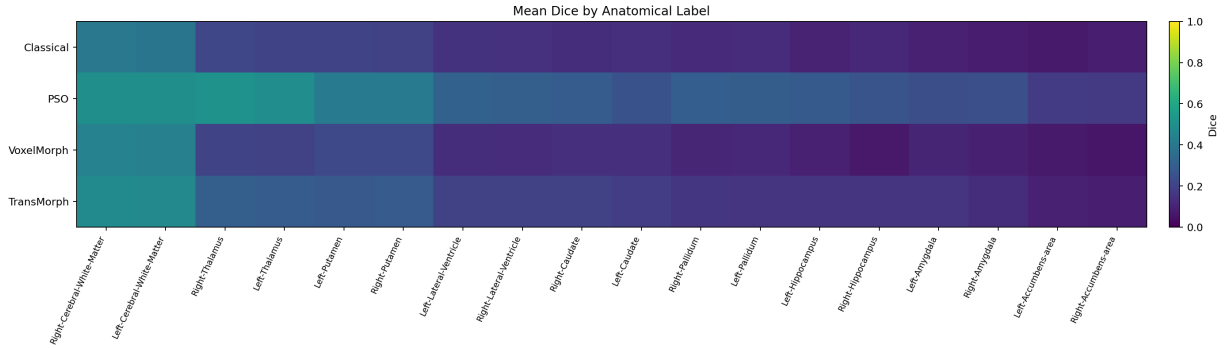


Figure 6.6: Mean Dice after registration for selected FreeSurfer anatomical labels.

Large structures such as cerebral white matter generally obtain higher Dice than small structures such as the amygdala or accumbens area. Downsampling to 64^3 has a stronger effect on small labels because only a few voxels may represent the structure. The heatmap also shows that one overall mean can hide large differences between anatomical regions.

Small labels are also more sensitive to interpolation and boundary errors. A shift of only a few voxels may have a minor effect on a large cortical or white matter region, but the same shift can substantially reduce Dice for a compact subcortical structure. For this reason, the label-level heatmap is a useful complement to the global mean Dice: it reveals whether a method improves anatomy broadly or mainly benefits large, easy-to-overlap structures.

6.7 Training Behavior

Table 6.3: Training summary for learned methods.

Method	Epochs	Training pairs	Best loss	Training time (s)
VoxelMorph	20	297	0.010429	421.9039
TransMorph	20	297	0.010865	639.8388

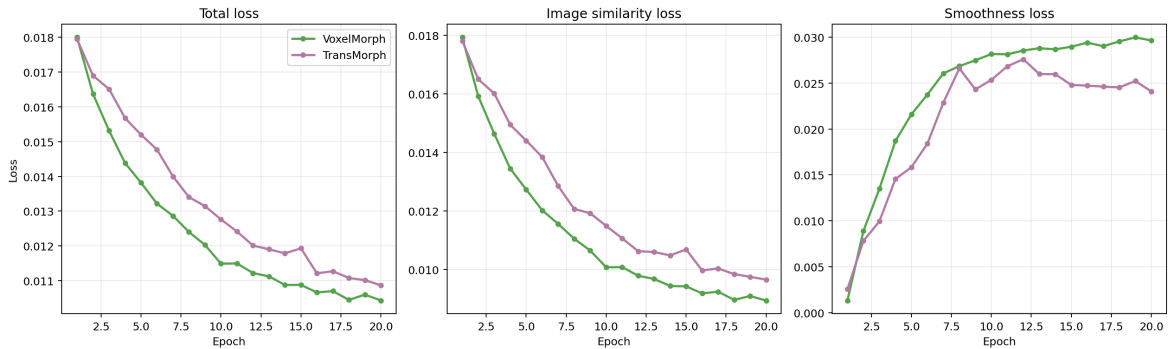


Figure 6.7: Training loss, image loss, and smoothness loss over 20 epochs.

Both models train without numerical failure. VoxelMorph reaches a slightly lower best loss and trains faster. TransMorph takes longer because attention and token processing add computation. Twenty epochs are enough for a functional benchmark, but they do not establish full convergence or reproduce the schedules used in the original papers.

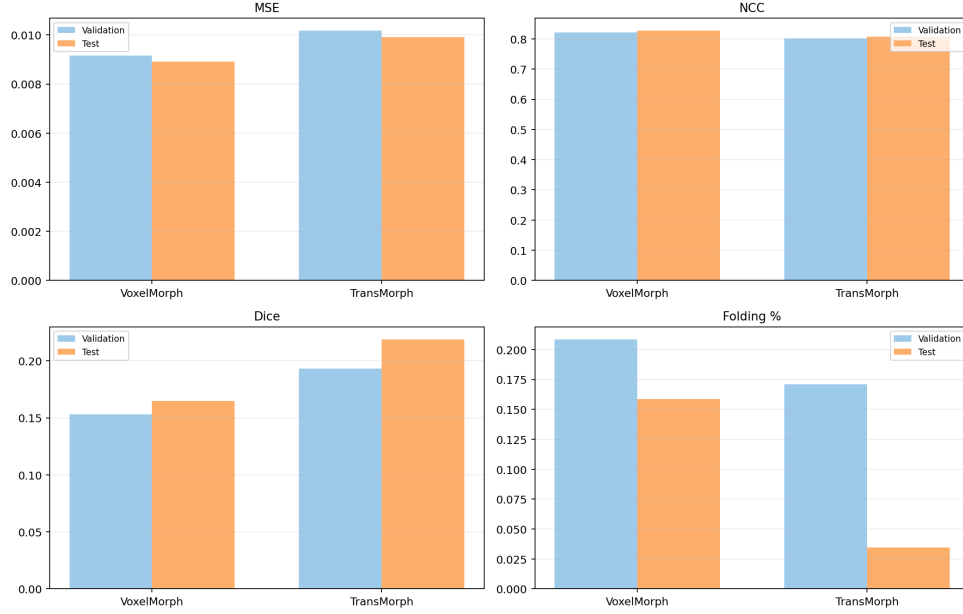


Figure 6.8: Validation and test comparison for VoxelMorph and TransMorph.

The validation and test results follow a similar trend, which suggests that the learned models are not failing only on the held-out test set. However, possible subject overlap between pair-level splits limits how strongly this can be interpreted as generalization to unseen subjects.

6.8 Qualitative Registration Example

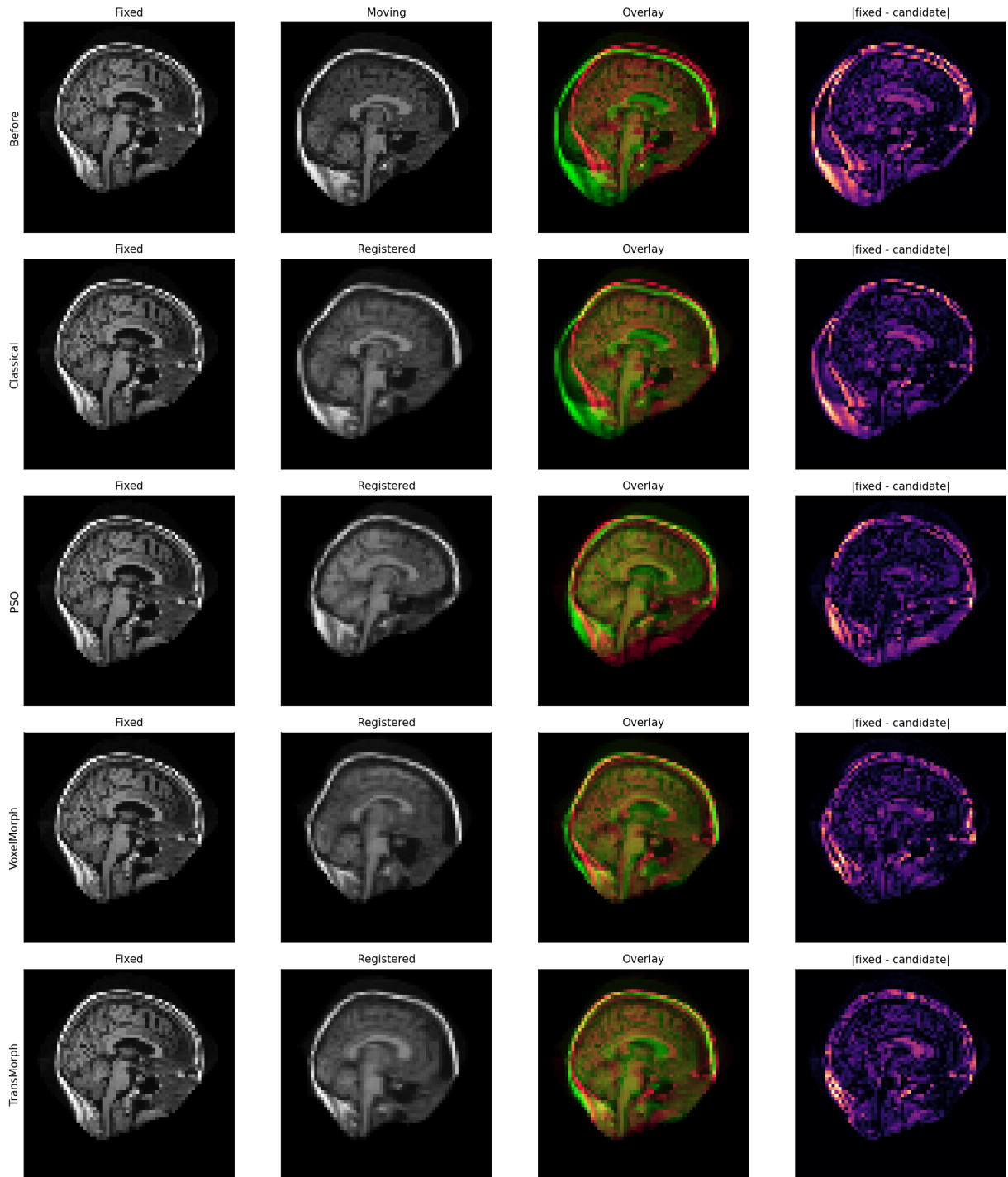


Figure 6.9: Qualitative comparison for a representative test pair. Rows compare the unregistered image and outputs from the four methods using registered slices, overlays, and absolute difference maps.

The qualitative figure is useful because a single metric cannot show where the remaining error is located. Red–green overlay disagreement highlights shifted edges, while the absolute difference map emphasizes intensity mismatch. The learned methods reduce

broad intensity differences, whereas PSO visibly changes global position and shape. Small local differences remain for every method.

6.9 Deformation and Folding

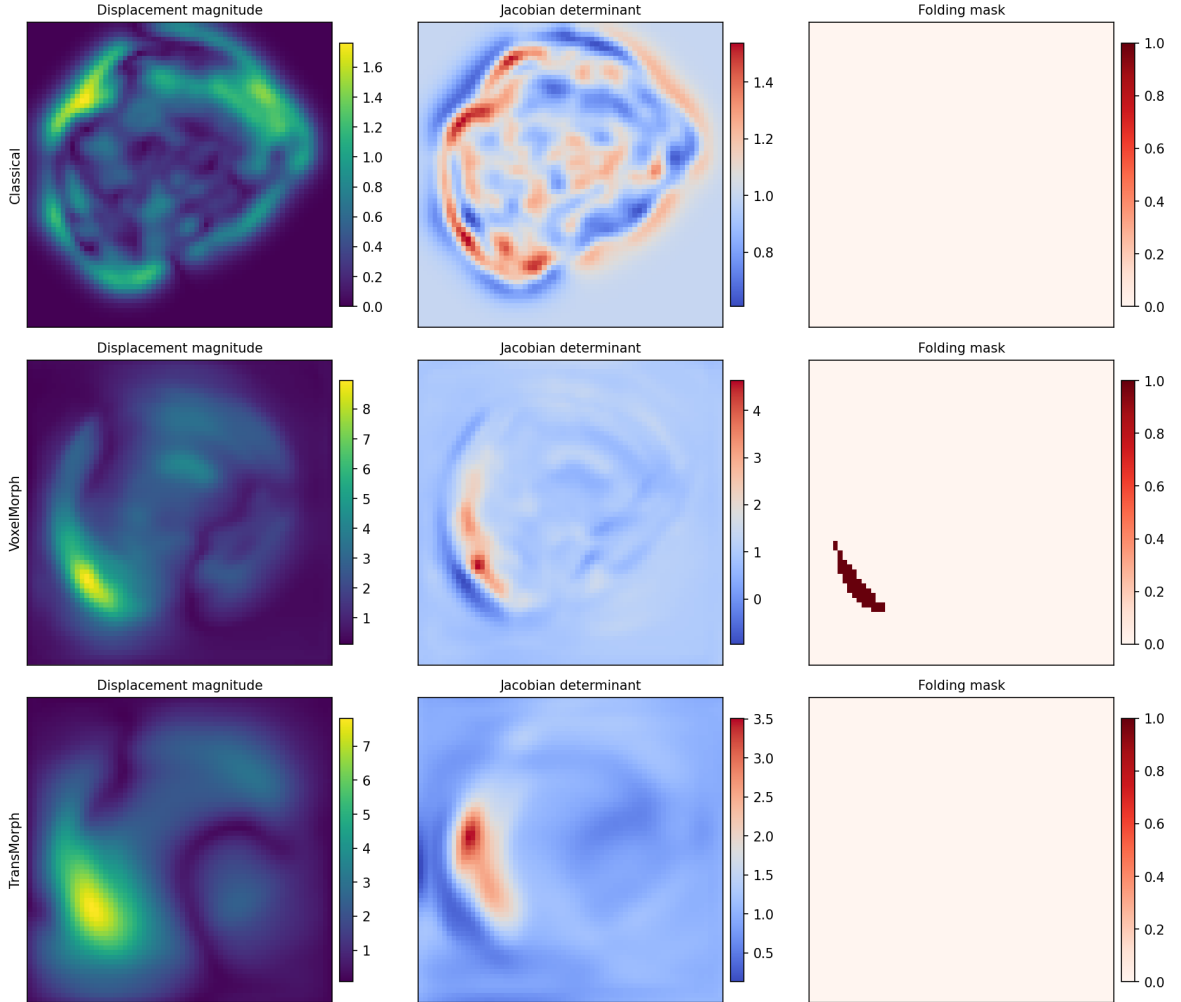


Figure 6.10: Displacement magnitude, Jacobian determinant, and folding mask for dense-field methods on a representative test pair.

Classical Demons has zero mean folding across the test set. TransMorph has a low folding percentage of 0.0346%, while VoxelMorph has 0.1585%. These values are small in absolute terms, but any fold can be undesirable when a transformation is used for anatomical measurement or label propagation. The result suggests that the compact TransMorph field is more regular than the compact VoxelMorph field under the same smoothness weight.

6.10 Animated Visualizations

The repository contains GIF visualizations for before/after comparison, axial slice sweeps, dense-field diagnostics, label contours, and PSO optimization progress. Standard

PDF reports cannot play GIF files, so representative static figures are used here. The full animation set is available at [visualize/oasis3d_benchmark_gifs](#).

It is important not to describe every animation as a true optimization iteration. PSO has genuine test-time iterations. Classical Demons is also iterative, although the stored visualization may use repeated runs with increasing iteration limits. VoxelMorph and TransMorph predict their final fields in one forward pass; scaling their final field from zero to full strength is only a visualization proxy, not a record of model optimization.

Chapter 7

Discussion

7.1 Why VoxelMorph Leads Intensity Metrics

VoxelMorph is trained directly with MSE image similarity. Its optimization target therefore closely matches the reported MSE metric. The CNN sees all 297 training pairs repeatedly and learns a fast mapping from local intensity patterns to local displacements. This explains its strong MSE and NCC results and its short inference time.

The result does not mean that VoxelMorph is always anatomically best. Its mean Dice is close to Classical and lower than PSO and TransMorph. It also has the highest folding percentage among dense methods. The model has learned to improve global intensity agreement, but this objective does not guarantee perfect boundaries for every FreeSurfer structure.

7.2 Why PSO Leads Dice

The FreeSurfer volumes are already broadly standardized, but residual global rotation, translation, scale, and shear differences remain. PSO explicitly searches these global affine components. Correcting a global mismatch can improve the overlap of many labels at the same time, which helps mean Dice.

PSO's advantage has a clear cost. Sixteen particles over 30 iterations require many volume resampling operations. The method is by far the slowest in the final table. It also cannot model local deformation. A possible future system would use PSO or another affine method for global pre-alignment and then apply a learned dense method for local refinement.

7.3 TransMorph as a Balanced Learned Method

TransMorph does not beat VoxelMorph on MSE or NCC, but it achieves higher Dice and lower folding. The Transformer token representation may help the model use broader context when predicting deformation. At the same time, the compact architecture remains fast at inference.

This behavior is consistent with the architectural difference between the two learned models. VoxelMorph’s small CNN emphasizes local convolutional matching, which is well suited to the MSE training objective and helps explain its strong intensity metrics. TransMorph first groups the volume into patch tokens and lets attention mix information across those tokens. Even in the compact one-layer version, this gives the model a different inductive bias: it can use broader spatial context when deciding how regions should move.

The result does not imply that attention automatically improves every metric. The early patch embedding can remove fine details, and the decoder must reconstruct a voxel-level field from coarse token features. This may explain why TransMorph is not the best intensity matcher in the experiment, even though its Dice and folding results are more favorable than the compact CNN.

This result should be interpreted as a comparison between the compact models in this repository, not a general conclusion that all Transformer registration models are better than all CNN models. Architecture size, training length, patch size, loss design, and preprocessing all affect the result.

7.4 Classical Stability

Classical Demons has zero measured folding and does not require a training set. It is therefore a useful reference method when deformation stability is important. Its mean runtime is still below one second at 64^3 , although this would increase at full resolution and with more iterations.

The zero-folding result should be read together with the algorithm’s conservative configuration. The smoothing parameter and low iteration count encourage gradual updates, which helps field regularity but may limit how far the method can move anatomical structures. In other words, the method is stable partly because it is not aggressively optimized for maximum overlap.

The baseline uses only 10 iterations, so it should not be interpreted as a fully tuned state-of-the-art Demons result. More iterations or a multi-resolution strategy could improve alignment, but would also change runtime and possibly field behavior.

7.5 Practical Method Selection

The results support different choices for different objectives:

- choose VoxelMorph when fast intensity alignment is the main goal;
- choose PSO affine when global anatomical overlap is more important than runtime;
- choose TransMorph when a learned dense method with better Dice and lower folding than the compact VoxelMorph is preferred;
- choose Classical Demons as a training-free and stable dense baseline.

In a clinical system, these criteria should be supplemented with landmark error, task-specific validation, uncertainty analysis, and expert review. A high average metric alone is not sufficient to establish clinical safety.

7.6 Comparison with Original Papers

The project implements the main ideas of VoxelMorph and TransMorph but does not exactly reproduce either paper. The original VoxelMorph work explores larger architectures, different losses, and broader experiments [12]. The original TransMorph uses a more complete hierarchical Swin Transformer design and paper-specific training choices [13].

The project models are intentionally compact: VoxelMorph starts with four channels, and TransMorph uses one Transformer encoder layer with embedding size 24. They are trained for only 20 epochs on 64^3 volumes. Consequently, the reported numbers should be used to compare the four local implementations under one protocol, not to compare directly with published paper tables.

7.7 Limitations

The main limitations are:

1. **Pair-level leakage risk.** Adjacent pairs share subjects, while splitting is performed after pair construction.
2. **Low spatial resolution.** Resizing to 64^3 removes fine detail and particularly affects small labels.
3. **Simplified geometry.** Prepared files do not fully preserve the original physical spacing and orientation.
4. **Compact learned models.** VoxelMorph and TransMorph are minimal implementations rather than exact reproductions.
5. **Limited training and tuning.** Twenty epochs and one main hyperparameter configuration are used.
6. **Single dataset and modality.** Only T1-weighted OASIS-1 brain MRI is evaluated in the final benchmark.
7. **No landmark ground truth.** Dice uses FreeSurfer segmentations, but no manually verified dense deformation or landmark set is available.
8. **Runtime scope.** Runtime is measured in this implementation and environment; it is not a hardware-independent complexity measure.

7.8 Future Work

The highest-priority improvement is a subject-level split. Unique subjects should be assigned to train, validation, and test sets before pairs are created. A second improvement is to preserve physical spacing and orientation and train at higher resolution using multi-resolution sampling.

Method improvements include affine pre-alignment followed by dense refinement, local NCC or mutual-information losses, inverse-consistency constraints, diffeomorphic integration, and explicit Jacobian penalties. Learned methods could be trained for longer

with data augmentation and validation-based checkpoint selection. A larger study should evaluate OASIS with official preprocessing and add a second dataset such as DIR-Lab, where landmark-based target registration error can be measured [15, 16].

Chapter 8

Conclusion

This project studied biomedical image registration from both a conceptual and an implementation perspective. Four representative approaches were implemented in a single benchmark: Classical Diffeomorphic Demons, affine Particle Swarm Optimization, a compact VoxelMorph-style CNN, and a compact TransMorph-style Transformer. All methods were evaluated on the same 85 OASIS-1 FreeSurfer-derived 3D test pairs, using the same prepared images, labels, runtime measurement procedure, and output format. This common protocol makes the comparison more meaningful than evaluating each method under a separate pipeline.

The main result is that registration performance is multi-dimensional. No method wins every metric because each method optimizes a different kind of alignment and has a different computational structure. MSE and NCC measure intensity agreement, Dice measures anatomical label overlap, runtime measures practical cost, and the Jacobian determinant measures deformation regularity. A high score in one metric therefore cannot be interpreted as complete registration quality. In this benchmark, VoxelMorph gives the best intensity alignment and the fastest mean runtime, PSO gives the best anatomical Dice, TransMorph provides a stronger balance between learned dense alignment and deformation stability, and Classical Demons gives the most conservative dense field with no detected folding.

VoxelMorph is the best choice when the main requirement is fast mono-modal intensity alignment. It achieves the lowest MSE, highest NCC, and fastest average runtime among the four methods. This is expected because the network is trained with an MSE-based unsupervised objective and predicts the deformation in one forward pass. For large-scale processing where many image pairs must be registered quickly and intensity similarity is the main target, the compact VoxelMorph model is the most efficient option in this project. However, its lower Dice and higher folding percentage show that good intensity matching does not automatically imply best anatomical boundary alignment or best deformation regularity.

PSO affine is the best choice when the goal is to maximize anatomical Dice in this benchmark and runtime is less important. It searches directly over global rotation, translation, scale, and shear parameters, which can improve the overlap of many FreeSurfer labels at once. This explains why it obtains the highest mean Dice even though it is not a dense deformable method. Its limitation is clear: it is much slower because many

candidate affine transforms must be evaluated for each pair, and it cannot model local anatomical deformation. Therefore, PSO is not the best standalone method for fast large-scale deformable registration, but it is a strong global pre-alignment component.

TransMorph is the most balanced learned dense method in this comparison. It does not surpass VoxelMorph on MSE or NCC, but it improves anatomical Dice and reduces folding relative to the compact CNN. This suggests that even a small Transformer-style token representation can provide useful broader spatial context when predicting deformation. If one learned dense method must be selected for a balanced trade-off between speed, anatomical overlap, and deformation stability, TransMorph is the more reasonable choice in this benchmark. This conclusion is limited to the compact implementation used here; it should not be interpreted as a universal claim that every Transformer registration model is better than every CNN registration model.

Classical Diffeomorphic Demons remains important as a stable and training-free baseline. It does not require a training set, does not depend on learned model parameters, and produces zero measured folding in the final test set. This makes it useful when deformation regularity and reproducibility are more important than maximizing Dice or intensity metrics. Its conservative behavior is also a limitation: with only 10 iterations, it may not move anatomy enough to obtain the best overlap. More iterations or a multi-resolution strategy could improve its accuracy, but would also change runtime and possibly deformation behavior.

Overall, the recommended method depends on the intended use:

- choose **VoxelMorph** for high-throughput intensity alignment, where speed, low MSE, and high NCC are the main priorities;
- choose **PSO affine** when global anatomical overlap is the main target and the additional runtime is acceptable;
- choose **TransMorph** when a learned dense method with a better balance of Dice, speed, and folding behavior is preferred;
- choose **Classical Demons** as a stable, training-free dense baseline or when deformation regularity is the central concern.

If the objective is to design a stronger practical registration pipeline rather than choose only one standalone method, the most promising direction is a hybrid workflow. A global affine step can first correct large pose and scale differences, and a learned dense method can then refine local anatomical deformation. In this project’s setting, that would mean using PSO or a faster affine optimizer for pre-alignment, followed by TransMorph or VoxelMorph for dense refinement. Such a pipeline would combine the strongest property of PSO, namely global anatomical alignment, with the strongest property of learned models, namely fast dense-field prediction.

Finally, the results should be interpreted within the limits of the experiment. The data are downsampled to 64^3 , the split is pair-level rather than subject-level, the learned models are intentionally compact, and Dice is computed from FreeSurfer segmentations rather than manually verified landmarks. The numbers therefore compare the four local implementations under one controlled protocol; they should not be read as definitive state-of-the-art results for the original VoxelMorph or TransMorph papers. Future work

should use subject-level splitting, higher-resolution preprocessing, stronger affine initialization, longer training, local NCC or mutual-information losses, inverse-consistency or diffeomorphic constraints, and landmark-based evaluation on additional datasets such as DIR-Lab. These improvements would make the benchmark more clinically meaningful and would better test whether the observed trade-offs remain stable across datasets and anatomical settings.

Appendix A

Selected FreeSurfer Labels

Table A.1: FreeSurfer labels used for Dice evaluation.

ID	Hemisphere	Structure
2	Left	Cerebral white matter
3	Left	Cerebral cortex
4	Left	Lateral ventricle
10	Left	Thalamus
11	Left	Caudate
12	Left	Putamen
13	Left	Pallidum
17	Left	Hippocampus
18	Left	Amygdala
26	Left	Accumbens area
41	Right	Cerebral white matter
42	Right	Cerebral cortex
43	Right	Lateral ventricle
49	Right	Thalamus
50	Right	Caudate
51	Right	Putamen
52	Right	Pallidum
53	Right	Hippocampus
54	Right	Amygdala
58	Right	Accumbens area

Appendix B

Repository Structure and Commands

```
src/  
  data/  
  methods/  
    classical/  
    metaheuristic/  
    voxelmorph/  
    transmorph/  
  utils/  
scripts/  
configs/  
tests/  
outputs/  
visualize/  
report_assets/
```

The main standalone commands are:

```
python -m src.run_method --method classical --fixed FIXED --moving MOVING --out  
  OUT  
python -m src.run_method --method pso --fixed FIXED --moving MOVING --out OUT  
python -m src.methods.voxelmorph.train --config configs/voxelmorph.yaml  
python -m src.methods.transmorph.train --config configs/transmorph.yaml  
python -m src.run_method --method voxelmorph --fixed FIXED --moving MOVING \  
  --checkpoint CHECKPOINT --out OUT  
python -m src.run_method --method transmorph --fixed FIXED --moving MOVING \  
  --checkpoint CHECKPOINT --out OUT
```

Appendix C

Team Contributions

Table C.1: Team contribution statement.

Name	Task	Contribution
Pham Cong Hoang	Leader, slide preparation, report writing, figure visualization, VoxelMorph implementation	25%
Le Tien Hop	Data processing, classical method implementation, report writing	25%
Tran Phong Quan	TransMorph implementation, report writing, literature survey	25%
Luu Hieu An	PSO implementation, figure visualization	25%

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